Gokul Sanjay Reddy Chatrala

B.Tech in Data Science & Artificial Intelligence • IIT Bhilai

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EDUCATION

Indian Institute of Technology (IIT) Bhilai — Raipur, Chhattisgarh, India

Bachelor of Technology (B.Tech) in Data Science and Artificial Intelligence — Expected Graduation: 2029

Current CGPA: 8.65 / 10.0 • *2nd Year Undergraduate*

TECHNICAL SKILLS

Languages:	Python, JavaScript, SQL, HTML5, CSS3
ML & Data Science:	scikit-learn, NumPy, Pandas, Matplotlib, Plotly, Seaborn, Streamlit
Frameworks & Web:	FastAPI, Flask, React, Tailwind CSS, SQLAlchemy, SQLite
Developer Tools:	Git, GitHub, VS Code, REST APIs, Linux/Bash

FEATURED PROJECTS

Crop Recommendation System | *Python, FastAPI, React, Tailwind, scikit-learn, SQLite, SQLAlchemy* — GitHub Repo ↗

- Built an end-to-end decision intelligence platform recommending the **top-3 most profitable crops** given soil chemistry, geographical climate, and prior crop season yields.
- Implemented a **two-stage ML pipeline** combining a RandomForest soil-suitability classifier with gradient-boosted yield and price regressors, factoring in crop-rotation penalties and personalized farmer profiles.
- Architected a modular **FastAPI backend** backed by SQLite/SQLAlchemy ORM, integrated with a responsive single-page **React + Tailwind CSS** user interface and reproducible synthetic data generation.

AutoInsight — Automated EDA Application | *Python, Streamlit, Pandas, Matplotlib, Plotly, Seaborn* — GitHub Repo ↗

- Engineered an automated Exploratory Data Analysis (EDA) web application that parses arbitrary CSV datasets with **automatic column-type inference** and comprehensive metadata profiling.
- Identified data anomalies including missing values, duplicate entries, and cardinality distributions; constructed six tabbed analytical views pairing static and interactive **Plotly visualisations**.
- Enabled instantaneous generation of downloadable diagnostic reports summarizing key trends and statistical distributions.

Concepts of Physics | *Python, Flask, HTML5, CSS3, JavaScript* — GitHub Repo ↗

- Developed an interactive educational portal providing intuitive physics learning modules and visual explanations to enhance conceptual understanding among undergraduate students.
- Constructed lightweight backend routing with Python (Flask) alongside custom responsive layouts.

ACADEMIC FOCUS & INTERESTS

Core Focus: Machine Learning Systems, Data Engineering, Workflow & Task Automation, Predictive Analytics, Full-Stack Prototyping.

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Multivariable Calculus, Probability & Statistics, Database Systems.